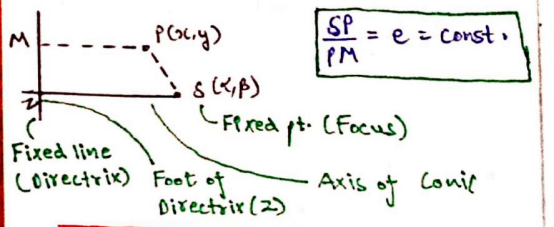
Parabola-1

#General Conic Section :



$\frac{SP}{PM} = e = \text{const.}$

① $(x-\alpha)^2 + (y-\beta)^2 = e^2\left(\frac{|lx+my+n|}{\sqrt{l^2+m^2}}\right)^2$

② $ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$

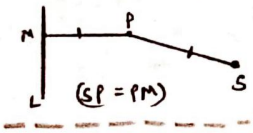
③ $\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$ → Tells about position of focus w.r.t directrix

$h^2 - ab$ → Tells about eccent. of the conic

④
- $\Delta = 0$ (Focus lie on Directrix)
  - $h^2-ab > 0$ ($e > 1$) ⇒ Pair of distinct line
  - $h^2-ab = 0$ ($e = 1$) ⇒ Pair of Coincident line
  - $h^2-ab < 0$ ($e < 1$) ⇒ Pair of Imag. line
- $\Delta \neq 0$ (Focus does not lie on directrix)
  - $h^2-ab > 0$ ($e > 1$) ⇒ Hyperbola
  - $h^2-ab = 0$ ($e = 1$) ⇒ Parabola
  - $h^2-ab < 0$ ($0 < e < 1$) ⇒ Ellipse
  - $h = 0, a = b$ ($e = 0$) ⇒ Circle

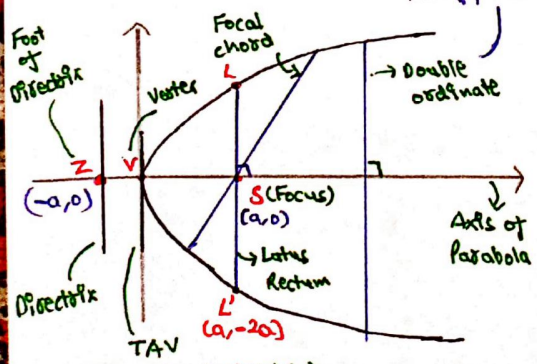
#Parabola :

Locus of a point P, which moves in a plane s.t. its distance from a fixed pt (S) & its distance from a fixed line (L) are equal.



$\left[e = \frac{SP}{PM} = 1\right]$

(Chord ⊥ to axis of parabola)



Here, $SZ = 2a$ and $SV = VS = a$

Standard Eqⁿ of Parabola :→

$y^2 = 4ax$

- $a > 0$
- $\alpha > 0$

] → Essential simultaneously for y^2 to always be positive

→ Parabola is symmetric about its axis.

Ends & length of Latus Rectum :→

Ends of LR

$L = (a, 2a)$ & $L' = (a, -2a)$

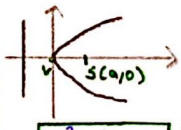
Length of LR $= 4a$

↳ Jo bhi parabola ki eqⁿ mein 'x' ke aage hota hai (coeff.), wo LR ki length hoti hai.

→ LR is the characteristic property of a Parabola.

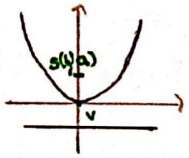
→ Two Parabola of equal length of LR and having same ends of LR would overlap on each other.

#Four Standard Parabolas : ($a > 0$)



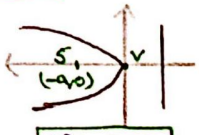
$y^2 = 4ax$

Focus ⇒ $S(a,0)$
Vertex ⇒ $(0,0)$
Directrix ⇒ $x = -a$
Axis ⇒ $y = 0$
Length of LR ⇒ $4a$
Ends of LR ⇒ $(a, \pm 2a)$
TAV ⇒ $x = 0$



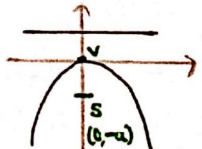
$x^2 = 4ay$

Focus ⇒ $S(0,a)$
Vertex ⇒ $(0,0)$
Directrix ⇒ $y = -a$
Axis ⇒ $x = 0$



$y^2 = -4ax$

Focus ⇒ $S(-a,0)$
Vertex ⇒ $(0,0)$
Directrix ⇒ $x = a$
Axis ⇒ $y = 0$
Length of LR ⇒ $4a$
Ends of LR ⇒ $(-a, \pm 2a)$
TAV ⇒ $x = 0$



$x^2 = -4ay$

Focus ⇒ $S(0,-a)$
Vertex ⇒ $(0,0)$
Directrix ⇒ $y = a$
Axis ⇒ $x = 0$

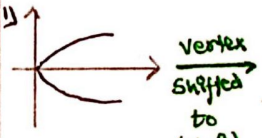
Length of LR ⇒ $4a$
Ends of LR ⇒ $(\pm 2a, a)$
TAV ⇒ $y = 0$

Length of LR ⇒ $4a$
Ends of LR ⇒ $(\pm 2a, -a)$
TAV ⇒ $y = 0$

#Parabola with shifted Vertex :

→ Here, the TAV and axis of parabola remain parallel to Coordinate Axes.

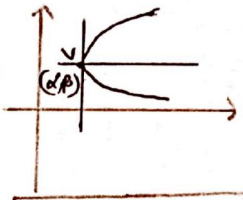
1)



$y^2 = 4ax$

$(y-0)^2 = 4a(x-0)^2$

Aise socho

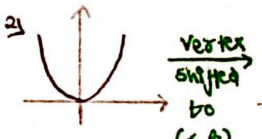


$(y-\beta)^2 = 4a(x-\alpha)^2$

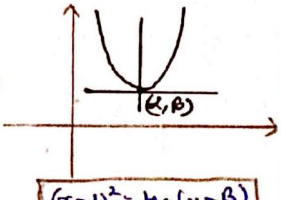
Eqⁿ Format : $x = Ay^2 + By + C$

(Quadratic in y)

2)



$(x-0)^2 = 4a(y-0)$



$(x-\alpha)^2 = 4a(y-\beta)$

Quadratic in x ←

Eqⁿ Format : $y = Ax^2 + Bx + C$

#Important Points :

(i) Length of LR = 2 (Distance b/w pts focus & directrix)
$= 2(SZ) = 2(2a) = 4a$

(ii) Vertex (V) is midpoint of Focus (S) & Foot of directrix (Z).

(iii) Length of LR = 4 (Dist. b/w focus & TAV)
= 4 (Dist. b/w focus & Vertex)
$= 4(SV) = 4(a)$

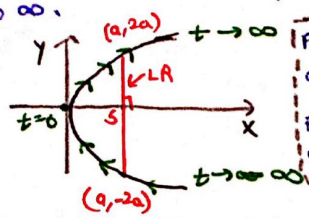
#Parametric Eqⁿ of Parabola :

While deciding parametric point give.

- (at^2) → to linear variable like x or y
- $(2at)$ → to second degree variable like x^2 or y^2

↳ Put minus (−) sign before at^2 accordingly, to maintain equality of given eqⁿ of parabola, after substituting parametric point

→ Parameter 't' varies from $-\infty$ to ∞.



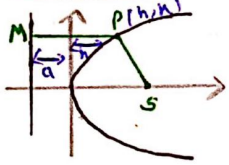
For Upper end of LR, $t=1$

For Lower end of LR, $t=-1$

Parabola	Parametric Coordinates
$y^2=4ax$	$(at^2, 2at)$
$y^2=-4ax$	$(-at^2, 2at)$
$x^2=4ay$	$(2at, at^2)$
$x^2=-4ay$	$(2at, -at^2)$
$(y-\beta)^2=4a(x-\alpha)$	$(at^2+\alpha, 2at+\beta)$
$(x-\alpha)^2=4a(y-\beta)$	$(2at+\alpha, at^2+\beta)$

Focal Distance :

→ By focus-directrix property, we know that $SP = PM$



So, we can write focal distance

$SP = PM = a+h$

$\left(a + \text{'x-coordinate of pt P'}\right)$

→ On the same line of thought, we can write focal distances as.

1) $y^2=4ax \;/\; y^2=-4ax \Rightarrow SP = a+|h|$

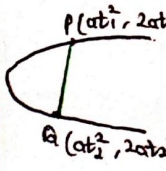
2) $x^2=4ay \;/\; x^2=-4ay \Rightarrow SP = a+|k|$

3) $(y-\beta)^2 = \pm 4a(x-\alpha) \Rightarrow SP = a+|h-\alpha|$

4) $(x-\alpha)^2 = \pm 4a(y-\beta) \Rightarrow SP = a+|k-\beta|$

Eqn of Chord joining two points :

→ We are discussing only for $y^2=4ax$



→ Write eqn of line in two point form for PQ

→ After simplifying, we get

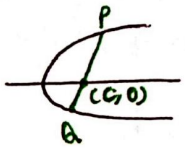
$$PQ: 2x - (t_1+t_2)y + 2at_1t_2 = 0$$

↳ Slope of this line is $m = \frac{2}{t_1+t_2}$

1) If chord PQ passes through $(c,0)$ on axis, then

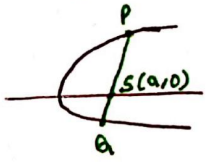
$2c + 2at_1t_2 = 0$

$\Rightarrow t_1t_2 = -c/a$



2) If PQ is a focal chord, then $c=a$ and

$t_1t_2 = -1$



Length of Focal Chord :

→ Discussion only for $y^2=4ax$



Since, $SP = PM$

$SQ = QM$

$PQ = SP + SQ$

$= a + x_P + a + x_Q$

$= a + at_1^2 + a + at_2^2$

$\Rightarrow PQ = a(t_1-t_2)^2$

$\Rightarrow PQ = a\left(t_1 + \frac{1}{t_1}\right)^2$ ($\because t_1t_2=-1$ for focal chord)

→ We know that in any real no., when its reciprocal is added

$\left(t_1 + \frac{1}{t_1}\right) \geq 2$ or $\left(t_1 + \frac{1}{t_1}\right) \leq -2$

$\Rightarrow \left(t_1 + \frac{1}{t_1}\right)^2 \geq 4$

$a\left(t_1 + 1/t_1\right)^2 \geq 4a$

Imp: Smallest possible focal chord is LR of parabola.

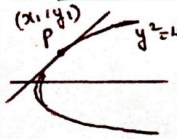
Important Point ⇒

Semi latus rectum of parabola $y^2=4ax$ is the harmonic mean of segments of focal chord of parabola.

i.e. $\frac{(2a)}{2} = \frac{1}{SP} + \frac{1}{SQ}$

Tangent to $y^2=4ax$:

(I) Cartesian Form ⇒



$T_P: yy_1 = 2a(x+x_1)$

$y^2 \to yy_1$; $x^2 \to xx_1$

$2x \to x+x_1$

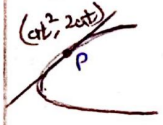
$2y \to y+y_1$

→ From above understanding, we can write eqn of tangent in cartesian form for any parabola.

Eg; $x^2=4ay$

$\Rightarrow T_P: xx_1 = 2a(y+y_1)$

(II) Parametric Form ⇒



Just put parametric coordinates in cartesian form of tangent.

$T_P: 2yat = 2a(x+at^2)$

$\Rightarrow T_P: ty = x+at^2$

Slope of tangent at P(t) is $(1/t)$

$m = \frac{1}{t}$

→ P.O.I of two tangents at t_1 & t_2 is $\left(at_1t_2,\ a(t_1+t_2)\right)$



→ Parametric form can be written for any type of parabola.

Parabola	Point of Contact	Tangent at t
$y^2=4ax$	$(at^2, 2at)$	$ty = x+at^2$
$y^2=-4ax$	$(-at^2, 2at)$	$ty = -x+at^2$
$x^2=4ay$	$(2at, at^2)$	$tx = y+at^2$
$x^2=-4ay$	$(2at, -at^2)$	$tx = -y+at^2$

(III) Slope form ⇒

Parabola	Point of Contact	Eqn of Tangent	Condn of Tangency
$y^2=4ax$	$\left(am^2, \frac{2a}{m}\right)$	$y = mx + \frac{a}{m}$	$c = am$
$y^2=-4ax$	$\left(\frac{-a}{m^2}, \frac{-2a}{m}\right)$	$y = mx - \frac{a}{m}$	$c = -am$
$x^2=4ay$	$(2am, am^2)$	$y = mx - am^2$	$c = -am^2$
$x^2=-4ay$	$(-2am, -am^2)$	$y = mx + am^2$	$c = am^2$

→ In case of non-standard parabola

$(y-\beta)^2 = 4a(x-\alpha)$

Tangent : $(y-\beta) = m(x-\alpha) + \frac{a}{m}$

Point of Contact : $\left(\alpha + \frac{a}{m^2},\ \beta + \frac{2a}{m}\right)$

Parabola-2

Common Tangent Questions :

1) $y = mx + 4$ is tangent on $y^2 = 4x$ and $x^2 = by$, find value of b.

Sol^n

$y = mx + 4$

$y^2 = 4x$: $4a = 4 \Rightarrow \boxed{a = 1}$

T_P : $y = mx + \frac{a}{m}$

i.e $\frac{a}{m} = 4$

$\Rightarrow \boxed{m = 1/4}$

$x^2 = by$: $4a = b \Rightarrow \boxed{a = b/4}$

T_P : $y = mx - am^2$

i.e $-am^2 = 4$

$-\frac{b}{4}\left(\frac{1}{4}\right)^2 = 4$

$\Rightarrow \boxed{b = -128}$ Ans

2) Find eqn of common tangent to $y^2 = 16x$ and $xy = -4$.

Sol^n Tangent at $y^2 = 16x$ $(a = 4)$

$y = mx + \frac{a}{m} \Rightarrow y = mx + \frac{4}{m}$

Now, whenever a line is tangent to a second degree curve, the quad. obtained from solving both eqn must have $D = 0$.

$y = mx + \frac{4}{m}$ $\quad xy = -4$

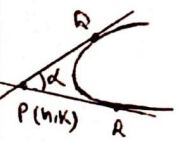
$x\left(mx + \frac{4}{m}\right) = -4 \Rightarrow m^2x^2 + 4x + 4m = 0$

$D = 16 - 16m^3 = 0 \Rightarrow \boxed{m = 1}$

So, $\boxed{T : y = x + 4}$ Ans

Tangent passing through a point (h, K) :

(Analysis valid only for $y^2 = 4ax$)



$y = mx + \frac{a}{m}$, passes through P

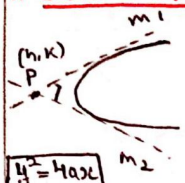
$\boxed{m_1 + m_2 = \frac{K}{h}}$ $\quad \boxed{m_1 m_2 = \frac{a}{h}}$

Two tangents of different slopes can be drawn from point P to the parabola

$\boxed{\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|}$ $\quad \boxed{\tan^2\theta = \frac{(m_1 + m_2)^2 - 4m_1 m_2}{(1 + m_1 m_2)^2}}$

Gives locus of point P(h,K) from which tangents to parabola enclose an angle of θ b/w them

Director Circle :



Locus of pts from which tangents are mutually ⊥ar.

$m_1 m_2 = -1 = \frac{a}{h}$

$\Rightarrow h = -a$ i.e $\boxed{x = -a}$

'DIRECTRIX' HI 'DIRECTOR CIRCLE' Hai

→ Two mutually ⊥ar tangents will always meet at the directrix of parabola.

Normal to the Parabola :

(I) Cartesian Form ⇒



$\frac{dy}{dx} = \frac{2a}{y_1} = m_T$

$m_T \cdot m_N = -1 \Rightarrow \boxed{m_N = \frac{-y_1}{2a}}$

$\boxed{N_P : y - y_1 = \frac{-y_1}{2a}(x - x_1)}$

↳ only for $y^2 = 4ax$

(II) Parametric Form ⇒

Put $P(at^2, 2at)$ in cartesian form

$N_P : (y - 2at) = \frac{-2at}{2a}(x - at^2)$

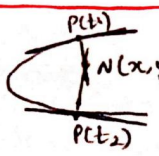
$\boxed{N_P : tx + y = 2at + at^3}$

Eqn of Parabola	Point of Contact	Normal at 't'
$y^2 = 4ax$	$(at^2, 2at)$	$y + tx = 2at + at^3$
$y^2 = -4ax$	$(-at^2, 2at)$	$y - tx = 2at + at^3$
$x^2 = 4ay$	$(2at, at^2)$	$xt + y = 2at + at^3$
$x^2 = -4ay$	$(2at, -at^2)$	$x - ty = 2at + at^3$

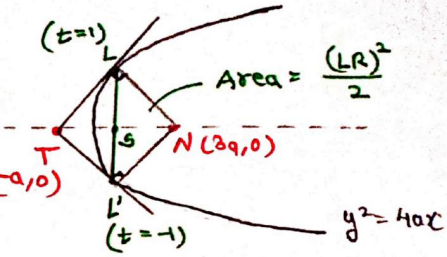
→ P.O.I of normals to parabola $y^2 = 4ax$ at $P(t_1)$ and $P(t_2)$ is

$\boxed{x = a(t_1^2 + t_2^2 + t_1t_2 + 2)}$

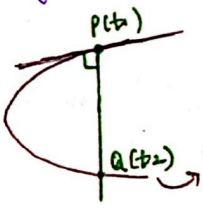
$\boxed{y = -at_1t_2(t_1 + t_2)}$



→ Tangents and Normals at ends of LR of a parabola form a square of $\boxed{\text{Area} = (LR)^2/2}$



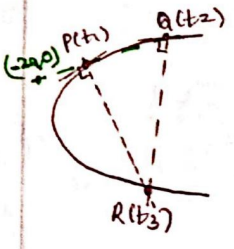
→ If the normals to the parabola $y^2 = 4ax$ at the point t_1 meets the parabola again at the point t_2, then



$\boxed{t_2 = -t_1 - \frac{2}{t_1}}$

Yeh Q pe necessarily normal nahi hai

→ If the normals to parabola $y^2 = 4ax$ at t_1 & t_2 intersect on the parabola at t_3, then



(i) $\boxed{t_1 t_2 = 2}$

(ii) $\boxed{t_1 + t_2 + t_3 = 0}$

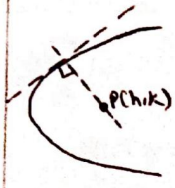
(iii) Line joining t_1 & t_2 passes through $\boxed{(-2a, 0)}$

(III) Slope Form ⇒

Parabola	Point of Contact	Eqn of Normal	Condn of Normality
$y^2 = 4ax$	$(am^2, -2am)$	$y = mx - 2am - am^3$	$c = -(2am + am^3)$
$y^2 = -4ax$	$(-am^2, 2am)$	$y = mx + 2am + am^3$	$c = 2am + am^3$
$x^2 = 4ay$	$\left(\frac{-2a}{m}, \frac{a}{m^2}\right)$	$y = mx + 2a + \frac{a}{m^2}$	$c = 2a + \frac{a}{m^2}$
$x^2 = -4ay$	$\left(\frac{2a}{m}, \frac{-a}{m^2}\right)$	$y = mx - 2a - \frac{a}{m^2}$	$c = -2a - \frac{a}{m^2}$

Normal passing through a Point (h, K) :

(Analysis only for $y^2 = 4ax$)



Upon solving for P(h,K) we get

$\boxed{am^3 + (2a - h)m + K = 0}$

(i) Max. 3 normals can be drawn through a point P(h,K)

(ii) Minimum 1 normal can be drawn through a point P(h,K).

(iii) If we are able to draw three (3) normals from $P(h,k)$ then

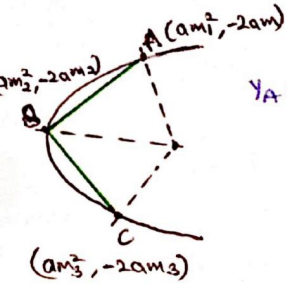
$$\boxed{h > 2a}$$ (Condⁿ on x coordinate)

(iv) $$\boxed{m_1 + m_2 + m_3 = 0}$$

$$\boxed{m_1m_2 + m_2m_3 + m_1m_3 = \frac{2a-h}{a}}$$

$$\boxed{m_1 \cdot m_2 \cdot m_3 = -\frac{k}{a}}$$

(v) A, B, C are conormal points & sum of their y-coordinates is zero.



$$y_A + y_B + y_C = -2a(m_1+m_2+m_3)$$
$$= -2a(0)$$
$$= 0$$

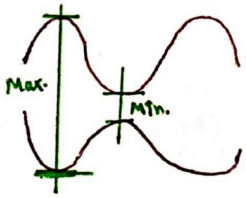
(vi) Centroid of $\triangle ABC$ lies on axis of the parabola.

$$\boxed{G = \left(\frac{a(m_1^2+m_2^2+m_3^2)}{3}, 0\right)}$$

→ Since $y=0$, lies on axis of parabola

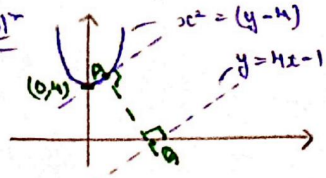
Shortest Distance :

Max. & min. distance b/w any 2 curves always occurs along common normal.



Eg: Find a point P on parabola $y^2 = x^2 + 4$ which is closest to line $y = 4x - 1$.

Soln



Slope of tangent at P & line would be same, as both have a common normal

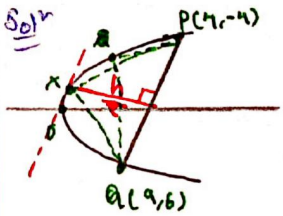
$y = x^2 + 4 \Rightarrow \frac{dy}{dx} = 2x = m_T$

& $m_T = 4$ (∵ Slope of line = 4)

$2x = 4 \Rightarrow \boxed{\begin{matrix} x = 2 \\ y = 8 \end{matrix}}$ Ans

Eg: $P(4,-4)$ & $Q(9,6)$ lie on $y^2 = 4x$ and let X be a point on arc POQ (O → Vertex) such that ar$(\triangle PXQ)$ is max. Find this area.

Soln



ar$(\triangle PXQ)$ →

$$= \frac{1}{2} \times PQ \times h$$

For area to be max, h must be max.

So, X would lie on arc POQ such that tangent at X is parallel to PQ.

$$m_{PQ} = \frac{10}{5} = 2 \quad ; \quad \frac{dy}{dx} = m_T = \frac{2}{y}$$

$$\frac{2}{y} = 2 \Rightarrow \boxed{y = 1}$$

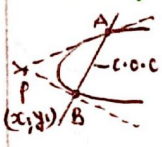
$$\boxed{x = 1/4}$$

So ar(PXQ) →

$$= \frac{1}{2}\begin{vmatrix} 1/4 & 1 & 1 \\ 4 & -4 & 1 \\ 9 & 6 & 1 \end{vmatrix} = \frac{75}{2}$$ Ans

Imp Equations (Same as circles) :

1) Chord of Contact ⇒



$$\boxed{T = 0}$$

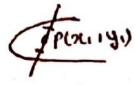
$$\boxed{yy_1 = 2a(x+x_1)}$$

2) Eqⁿ of Pair of Tangents ⇒

$$\boxed{SS_1 = T^2}$$

$$(y^2-4ax)(y_1^2-4ax_1) = \left[yy_1 - 2a(x+x_1)\right]^2$$

3) Eqⁿ of chord with given mid-point ⇒

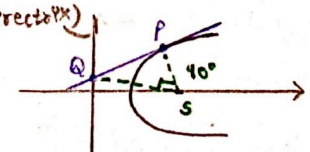


$$\boxed{T = S_1}$$

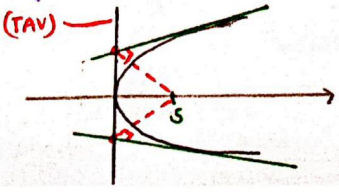
$$yy_1 - 2a(x+x_1) = y_1^2 - 4ax_1$$

Imp Highlights :

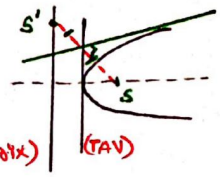
H-1 ⇒ The portion of tangent b/w directrix and the curve subtends 90° at focus.



H-2 ⇒ Foot of ⊥ar from focus on any tangent lies on TAV (Tangent at vertex).

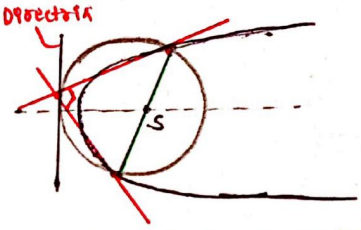


H-3 ⇒ Image of focus (S) in any tangent to the parabola lies on its directrix.

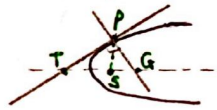


H-4 ⇒ Tangents at extremities of a Focal chord intersect at 90° on directrix

⇒ Hence, any circle drawn, taking a Focal chord as diameter always touches directrix.



H-5 ⇒ Tangent & Normal are drawn at a point P and focal distance of P is SP, then

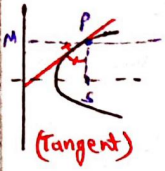


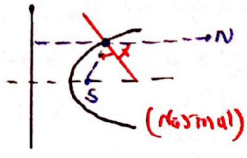
$$\boxed{ST = SG = SP}$$

Means, if a circle drawn taking TG as diameter, then it passes through P & has centre at Focus (S)

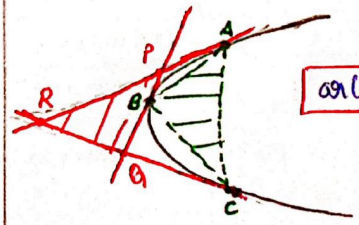
H-6 ⇒ (i) Tangent bisect angle b/w focal radius & ⊥ar from directrix.

(ii) Normal bisect angle b/w focal radius & ⊥ar from directrix.





H-7 ⇒ Area of Δ made by 3 points on a parabola is double of area of Δ formed by tangents at these 3 points.



$$\boxed{ar(ABC) = 2\,ar(PQR)}$$

H-8 ⇒ Also, orthocentre of $\triangle PQR$ lies on Directrix of parabola.

$$\boxed{H \equiv (-a,\ a(t_1+t_2+t_3+t_1t_2t_3))}$$